Nikhil Jain

+91-8619744807 nikhilj22.iitk@gmail.com
Bengaluru, India | GitHub | LinkedIn

Education

Indian Institute of Technology, Kanpur Oct 2022 - May 2026
B.Tech in Electrical Engineering *CPI: 8.0 / 10*
Minors in Computer Systems, Machine Learning and Algorithms
Key courses: Data Structures & Algorithms, Algorithms-2, Randomised Algorithms, Database Systems, Computer Networks, Statistical Natural Language Processing, Introduction to Machine Learning, Probability and Statistics, Communication Systems.

Experience

Squarepoint Capital, Bengaluru Jul 2026 - Sep 2026
Systems specialist, TradeFlowSupport Team

- Built a reusable **Bash/SSH diagnostics framework** for parallel host checks across hosts, cutting manual effort by **5**×.
- Triaged high-severity **Linux, Kubernetes, compute, storage, networking and application** incidents across NYC, LON and SGO data centers using logs, monitoring, CLI checks and host-health data to restore production service.
- Led bridges with **8+ trading teams** and infrastructure owners to isolate faults, restore service and drive remediation.
- Resolved **20+ break-fix** incidents at **Equinix** & **CoreSite**, coordinating vendors & validating recovery through health checks.

Samsung R&D Institute India - Bengaluru May 2025 - Jul 2025
Research Developer Intern, MAC/UL2 Team

- Mapped LTE/NR scheduling and retransmissions across **MAC, RLC and PDCP** layers to large-scale modem logs.
- Tuned **XGBoost** with grid search for UE–base-station modulation and coding scheme **(MCS) prediction** for **faulty nodes**.
- Reported **96.1% regression** and **92.8% classification directional accuracy** (CQI, SINR, RBs, power headroom).
- Found **UE anomalies** and **PDCP packet loss** in modem logs to inform MCS prediction and channel reliability.
- Reviewed **Deep RL for Link Adaptation (DRLLA)** for dynamic MCS selection to reduce BLER in 5G networks.

Projects

API Authorization Regression Lab Fall 2026
Personal project; local synthetic data

- Built a multi-tenant document API (roles, shares, nested comments, cursor paging, ETags) and a reference rules model with independent authorization checks; the clean service had **0** mismatches over ∼**56K** requests.
- Of **30** seeded bugs, happy-path tests caught 4, enumeration **24**, and model-guided sequences **30/30** across 10 seeds.
- Wrote a **shrinker** reducing failing random runs to **1**–**3**-step repro stories (median 14 steps → 1).

Founder & Developer, Sniff.in | 2026
Presence-based social discovery app; deployed backend and mobile test builds, React Native, TypeScript

- Built a **mutual-consent social app** for campuses, offices and events, with a deployed backend and iOS/Android test builds.
- Built **50+ React Native/TypeScript screens** with **chat, media, push notifications** and QR/invite onboarding.
- Designed **39 Postgres tables** with row-level security and **89 role-checked, rate-limited functions**; added consent-controlled pgvector matching, a deterministic fallback and email verification against 10,500+ university domains.
- Passed **125+ automated tests**: 69 database security tests as real user roles, 31 unit tests and 25 live-backend end-to-end checks. **Race-condition cases** exercised authorization, rate limits and concurrent state changes.

IncidentOps: Real-Time Incident Tracker Summer 2026
Personal full-stack project; Spring Boot, React, TypeScript, REST

- Built a **Spring Boot/JPA and React/TypeScript** incident tracker with an enforced **OPEN→ACK→RESOLVED state machine**, append-only audit timeline, RFC 7807 errors and live updates via Server-Sent Events.
- Added optimistic locking: in an 8-way ack race exactly **1 of 8** won (**6 of 8** without locking), across **50/50** rounds.
- Added severity-based SLAs, auto-escalation and on-call rotation; validated using **45** JUnit/MockMvc and **17** Vitest tests.

Technical Skills

Programming & Practices: C++, C, Python, Bash, SQL, data structures and algorithms, object-oriented programming, Git, CI/CD; familiarity with Spring Boot, REST APIs, pytest, software development lifecycle (SDLC)
Distributed Systems & Infrastructure: Linux, Docker, Kubernetes, distributed applications, Redis, Kafka, GCP, incident triage
Databases & Data: PostgreSQL, relational databases, query optimization, MongoDB, Supabase, NumPy, Pandas

Academic Achievements & Leadership

- **Samsung SWC Pro Test (2025)** Cleared; Received PPO Offer.
- **Google Developer Group (2025)** Campus Organiser 2025–26
- **ANC Research Wing Manager(2024)** Led 18 secretaries
- **ICPC** Regionalist (2026); Codeforces nikhilj22.iitk - **Expert**.
- **JEE Main 2022:** AIR 343; **JEE Advanced:** AIR 1240
- **KVPY:** AIR 41 (SA 2021), AIR 248 (SX 2022)
- **NTSE Scholarship (2020):** Awarded by NCERT